community. However, certain species, including deer or rabbits, may migrate or disperse to another community in the metacommunity, moving to a different patch of forest in search of better opportunities to feed, shelter, or breed. Differing types of habitat will influence this balance between interlinked and independent development. The theory of metacommunities provides a framework for studying how and why variations develop and their impact on biodiversity and population fluctuations.

Local versus regional

A major advantage of looking at communities in this spatial way is that it may help resolve a number of seemingly contradictory observations. One ecologist's study, for instance, might look at the way species live and interact together in a small local community. This narrowly focused study finds that competition between species for resources is a crucial factor in the workings of the community. Another study might look at the picture across a larger community. This macro-study discovers that competition plays virtually no part. So which result is correct? The answer may be that both are right, and the difference simply depends on scale. The benefit of metacommunity theory is that it allows ecologists to reconcile these differences. It enables them to look for explanations on both a local and regional scale.

A metacommunity might be a set of half a dozen deciduous trees within a park, with each tree an individual community. However, it could equally be all the deciduous forests in temperate zones all around the world. What metacommunity theory does is allow ecologists to work at any scale, at least in theory.